

# Discrete Structures

Spring 2026 – Week 4

# Logical Equivalence

## Logical Equivalence

- Two propositions  $p$  and  $q$  are **logically equivalent** if their truth tables are the same.
- Namely,  $p$  and  $q$  are **logically equivalent** if  $p \leftrightarrow q$  is a tautology. If  $p$  and  $q$  are logically equivalent, we write  $p \equiv q$ .

| $P$ | $Q$ | $P \leftrightarrow Q$ |
|-----|-----|-----------------------|
| T   | T   | T                     |
| T   | F   | F                     |
| F   | T   | F                     |
| F   | F   | T                     |

**\*\***

A tautology is a statement in logic that is always true, no matter whether its variables are true or false.

**Example:**  $p \vee \neg p$

(This is always true, because either  $p$  is true or it isn't)

### Example:

Look at the following two compound propositions:  $p \rightarrow q$  and  $q \vee \neg p$ .

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

| $p$ | $q$ | $\neg p$ | $q \vee \neg p$ |
|-----|-----|----------|-----------------|
| T   | T   | F        | T               |
| T   | F   | F        | F               |
| F   | T   | T        | T               |
| F   | F   | T        | T               |

- The last column of the two truth tables are identical. Therefore  $(p \rightarrow q)$  and  $(q \vee \neg p)$  are **logically equivalent**.
- So  $(p \rightarrow q) \leftrightarrow (q \vee \neg p)$  is a tautology.
- Thus:  $(p \rightarrow q) \equiv (q \vee \neg p)$ .

### Example:

By using truth table, prove  $p \oplus q \equiv \neg(p \leftrightarrow q)$ .

We have a number of rules for logical equivalence. For example:

De Morgan Law:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \quad (1)$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad (2)$$

The following is the truth table proof for (1). The proof for (2) is similar.

| $p$ | $q$ | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|-----|-----|--------------|--------------------|----------|----------|----------------------|
| F   | F   | F            | T                  | T        | T        | T                    |
| F   | T   | F            | T                  | T        | F        | T                    |
| T   | F   | F            | T                  | F        | T        | T                    |
| T   | T   | T            | F                  | F        | F        | F                    |

## Distributivity

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \quad (1)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r) \quad (2)$$

The following is the truth table proof of (1). The proof of (2) is similar.

| $p$ | $q$ | $r$ | $q \wedge r$ | $p \vee (q \wedge r)$ | $p \vee q$ | $p \vee r$ | $(p \vee q) \wedge (p \vee r)$ |
|-----|-----|-----|--------------|-----------------------|------------|------------|--------------------------------|
| T   | T   | T   | T            | T                     | T          | T          | T                              |
| T   | T   | F   | F            | T                     | T          | T          | T                              |
| T   | F   | T   | F            | T                     | T          | T          | T                              |
| T   | F   | F   | F            | T                     | T          | T          | T                              |
| F   | T   | T   | T            | T                     | T          | T          | T                              |
| F   | T   | F   | F            | F                     | T          | F          | F                              |
| F   | F   | T   | F            | F                     | F          | T          | F                              |
| F   | F   | F   | F            | F                     | F          | F          | F                              |

## Contrapositives

The proposition  $\neg q \rightarrow \neg p$  is called the **Contrapositive** of the proposition  $p \rightarrow q$ . They are logically equivalent.

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

| $p$ | $q$ | $\neg q \rightarrow \neg p$ |
|-----|-----|-----------------------------|
| T   | T   | T                           |
| T   | F   | F                           |
| F   | T   | T                           |
| F   | F   | T                           |

# Proving Logical Equivalence

- By using equivalence laws, we can prove two propositions are logically equivalent without having to construct large truth tables

## In General:

- Each line should list the law that led to it
  - Exactly one law applied per line
- Start with LHS and go until you reach the RHS



## Logical Equivalence using Laws

| Equivalenc                                   | Nam             |
|----------------------------------------------|-----------------|
| $p \wedge T \equiv p$<br>$p \vee F \equiv p$ | Identity Laws   |
| $p \vee T \equiv T$<br>$p \wedge F \equiv F$ | Domination Laws |
| $p \vee p \equiv p$<br>$p \wedge p \equiv p$ | Idempotent Laws |

## Logical Equivalence using Laws

| Equivalenc                                                                                   | Nam              |
|----------------------------------------------------------------------------------------------|------------------|
| $\neg(\neg p) \equiv p$                                                                      | Double Negation  |
| $p \vee \neg p \equiv T$<br>$p \wedge \neg p \equiv F$                                       | Negation Laws    |
| $\neg(p \vee q) \equiv \neg p \wedge \neg q$<br>$\neg(p \wedge q) \equiv \neg p \vee \neg q$ | De Morgan's Laws |
| $p \rightarrow q \equiv \neg p \vee q$                                                       | Implication Law  |
| $p \vee (p \wedge q) \equiv p$<br>$p \wedge (p \vee q) \equiv p$                             | Absorption Law   |

## Logical Equivalence using Laws

| Equivalenc                                                                                                               | Nam               |
|--------------------------------------------------------------------------------------------------------------------------|-------------------|
| $p \vee q \equiv q \vee p$<br>$p \wedge q \equiv q \wedge p$                                                             | Commutative Laws  |
| $(p \vee q) \vee r \equiv p \vee (q \vee r)$<br>$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$                     | Associative Laws  |
| $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$<br>$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | Distributive Laws |

**TABLE 6** Logical Equivalences.

| <i>Equivalence</i>                                                                                                       | <i>Name</i>         |
|--------------------------------------------------------------------------------------------------------------------------|---------------------|
| $p \wedge \mathbf{T} \equiv p$<br>$p \vee \mathbf{F} \equiv p$                                                           | Identity laws       |
| $p \vee \mathbf{T} \equiv \mathbf{T}$<br>$p \wedge \mathbf{F} \equiv \mathbf{F}$                                         | Domination laws     |
| $p \vee p \equiv p$<br>$p \wedge p \equiv p$                                                                             | Idempotent laws     |
| $\neg(\neg p) \equiv p$                                                                                                  | Double negation law |
| $p \vee q \equiv q \vee p$<br>$p \wedge q \equiv q \wedge p$                                                             | Commutative laws    |
| $(p \vee q) \vee r \equiv p \vee (q \vee r)$<br>$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$                     | Associative laws    |
| $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$<br>$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ | Distributive laws   |
| $\neg(p \wedge q) \equiv \neg p \vee \neg q$<br>$\neg(p \vee q) \equiv \neg p \wedge \neg q$                             | De Morgan's laws    |
| $p \vee (p \wedge q) \equiv p$<br>$p \wedge (p \vee q) \equiv p$                                                         | Absorption laws     |
| $p \vee \neg p \equiv \mathbf{T}$<br>$p \wedge \neg p \equiv \mathbf{F}$                                                 | Negation laws       |

**TABLE 7** Logical Equivalences  
Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

**TABLE 8** Logical  
Equivalences Involving  
Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

## Logical Equivalence using Laws

**Example:**  $\neg(p \vee (\neg p \wedge q))$  is logically equivalent to

**Show that**  $\neg p \wedge \neg q$

### **Solution:**

|                                  |          |                                                 |                                    |
|----------------------------------|----------|-------------------------------------------------|------------------------------------|
| $\neg(p \vee (\neg p \wedge q))$ | $\equiv$ | $\neg p \wedge \neg(\neg p \wedge q)$           | by the second De Morgan law        |
|                                  | $\equiv$ | $\neg p \wedge [\neg(\neg p) \vee \neg q]$      | by the first De Morgan law         |
|                                  | $\equiv$ | $\neg p \wedge (p \vee \neg q)$                 | by the double negation law         |
|                                  | $\equiv$ | $(\neg p \wedge p) \vee (\neg p \wedge \neg q)$ | by the second distributive law     |
|                                  | $\equiv$ | $F \vee (\neg p \wedge \neg q)$                 | because $\neg p \wedge p \equiv F$ |
|                                  | $\equiv$ | $(\neg p \wedge \neg q) \vee F$                 | by the commutative law             |
|                                  |          |                                                 | for disjunction                    |
|                                  | $\equiv$ | $(\neg p \wedge \neg q)$                        | by the identity law for <b>F</b>   |

## Logical Equivalence Using Laws

Is  $(q \wedge p) \vee \neg(q \rightarrow p) \equiv q$ ?

$$(q \wedge p) \vee \neg(q \rightarrow p)$$

$$\equiv (q \wedge p) \vee \neg(\neg q \vee p)$$

$$\equiv (q \wedge p) \vee (\neg(\neg q) \wedge \neg p)$$

$$\equiv (q \wedge p) \vee (q \wedge \neg p)$$

$$\equiv q \wedge (p \vee \neg p)$$

$$\equiv q \wedge T$$

$$\equiv q$$

Original

Implication Law

De Morgan's

Law

Double Negation

Law Distributive

Law Negation Law

Domination Law

## Logical Equivalence Using Laws

Is  $(P \rightarrow R) \vee (Q \rightarrow R) \equiv (P \wedge Q) \rightarrow R$ ?

$$(P \rightarrow R) \vee (Q \rightarrow R)$$

Original LHS

$$\equiv (\neg P \vee R) \vee (Q \rightarrow R)$$

Implication Law

$$\equiv (\neg P \vee R) \vee (\neg Q \vee R)$$

Implication Law

$$\equiv (\neg P \vee \neg Q) \vee (R \vee R)$$

Associative Law

$$\equiv (\neg P \vee \neg Q) \vee R$$

Idempotent Law

$$\equiv \neg(P \wedge Q) \vee R$$

DeMorgan's

$$\equiv (P \wedge Q) \rightarrow R$$

Law Implication

$$= \text{RHS}$$

Law



## Logical Equivalence Using Laws

### Exercise:

**Prove  $(p \rightarrow q) \wedge (p \rightarrow r) \equiv \neg p \vee (q \wedge r)$**

$$\begin{aligned}(p \rightarrow q) \wedge (p \rightarrow r) &\equiv (\neg p \vee q) \wedge (p \rightarrow r) && \text{by Conditional Law} \\ &\equiv (\neg p \vee q) \wedge (\neg p \vee r) && \text{by Conditional Law} \\ &\equiv \neg p \vee (q \wedge r) && \text{by Distributive Law}\end{aligned}$$

## Logical Equivalence Using Laws

**Use the logical laws to show the following are equivalent.**

$$1. \quad \neg(p \rightarrow \neg q) \equiv p \wedge q$$

$$2. \quad (\neg p \rightarrow (q \wedge \neg q)) \equiv p$$

## SOLUTION:

$$1. \quad \neg(p \rightarrow \neg q) \equiv p \wedge q$$

Start with the left-hand side:

Use implication law:  $p \rightarrow r \equiv \neg p \vee r$

$$\neg(\neg p \vee \neg q)$$

Apply De Morgan's Law:

$$p \wedge q$$

## SOLUTION:

$$2. (\neg p \rightarrow (q \wedge \neg q)) \equiv p$$

Start with the left-hand side

Use implication law:

$$\neg(\neg p) \vee (q \wedge \neg q)$$

Apply double negation:

$$p \vee (q \wedge \neg q)$$

Apply negation law  $q \wedge \neg q \equiv F$

$$p \vee F$$

Apply identity law:

$$p$$